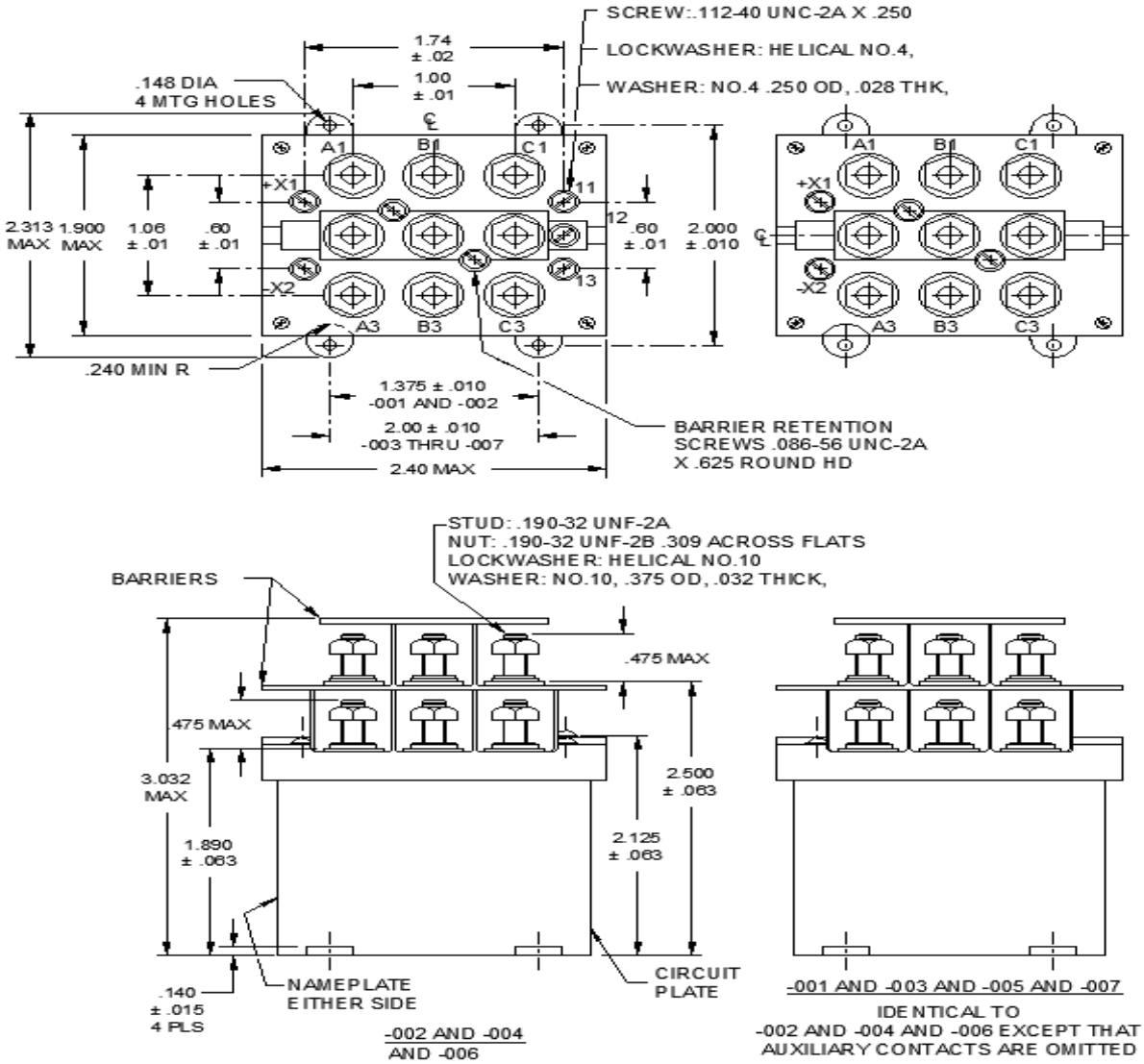
INCH-POUND

MIL-PRF-6106/9G
15 June 2022
SUPERSEDING
MIL-PRF-6106/9F
24 July 2020

PERFORMANCE SPECIFICATION SHEET

RELAY, ELECTROMAGNETIC, CONTINUOUS DUTY, 60 AMP, 3PDT, ENVIRONMENTALLY SEALED

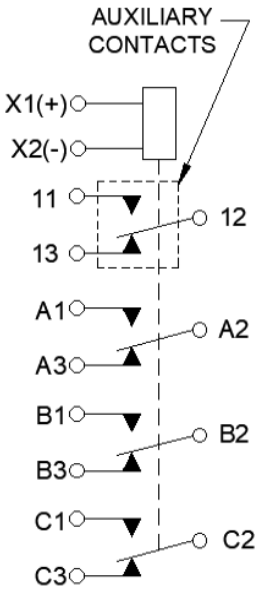
This specification is approved for use by all Departments and Agencies of the Department of Defense.
The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-PRF-6106.



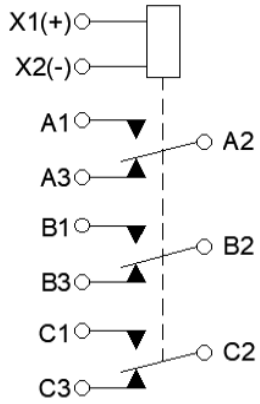
AMSC N/A
FSC 5945

DISTRIBUTION STATEMET A. Approved for public release; distribution is unlimited.

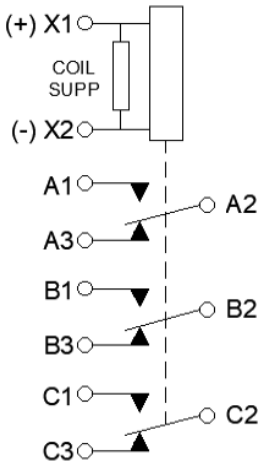
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CIRCUIT
-002 AND -004
AND -006



CIRCUIT
-001 AND -003
AND -005



CIRCUIT
-007

Inches	mm	Inches	mm	Inches	mm
.010	0.25	.309	7.86	1.890	48.01
.02	0.51	.375	9.53	1.900	48.26
.028	0.71	.440	11.18	2.000	50.80
.032	0.81	.600	15.24	2.115	53.72
.063	1.60	.625	15.88	2.313	58.75
.148	3.76	1.000	25.40	2.40	60.96
.186	4.72	1.060	26.92	2.500	63.50
.240	6.10	1.375	35.18	3.032	77.01
.250	6.35	1.74	44.20		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.031 (0.79 mm).
4. Weights include cover, barriers, and terminal hardware.
5. When using relay at rated 60 amps, the use of lugs per Burndy part number YAEV6C-G8 or equal is recommended.
6. Terminal temperature rise under continuous current conditions is 95°C maximum.

FIGURE 1. Relay, outline drawing – Continued.

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TABLE I. Dash number and operating characteristics.

Dash no.	Type	Coil	Nominal			Coil data maximum			Max Normal 1/	pickup High temp test	voltage 1/ Cont. current test	Hold voltage 1/	Drop out voltage 1/	Time Operate 2/	millisec Release 3/	onds Bounce Maintained		maximum Auxiliary	
			Volts	Freq	Res	Volts	Ampere Inrush	Ampere Steady state								NO	NC	NO	NC
-001 AND -005	1	X1, X2	28	DC	-	29	1.25	.25	17	20	21.5	7.0	1.5	25	25	2	2	-	-
-002 AND -006	1	X1, X2	28	DC	-	29	1.25	.25	17	20	21.5	7.0	1.5	25	25	2	2	2	2
-003	1	X1, X2	115	400	-	124	.71	.1	85	95	100	40	10	25	55	2	2	-	-
-004	1	X1, X2	115	400	-	124	.71	.1	85	95	100	40	10	25	55	2	2	2	2
-007 4/	1	X1, X2	28	DC	-	29	1.25	.25	17	20	21.5	7.0	1.5	25	25	2	2	-	-

1/ Over the temperature range 2/ With nominal coil voltage 3/ From nominal coil voltage
4/ -007 is same as -001 except for transient suppressed coil

TABLE II. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles $x10^3$	28 V dc 1/ Main NO	Main NC	Auxiliary NO	Auxiliary NC	115 V ac, 1 phase 1/ Main 400 Hz	Main 60 Hz	Auxiliary 400 Hz	Auxiliary 60 Hz	115/200 V ac, 3 phase 1/ Main 400 Hz	Main 60 Hz	Auxiliary 400 Hz	Auxiliary 60 Hz
Resistive	100	-	-	3	3	60	-	3	-	60	-	-	-
Inductive	100	-	-	1.5	1.5	60	-	1.5	-	60	-	-	-
Motor	50	-	-	-	-	40	-	-	-	40	-	-	-
Lamp		-	-	.5	.5		-	.5	-		-	-	-
Transfer load 2/		-	-				-		-		-	-	-
Mechanical life reduced current	200	-	-	.75	.75	15	-	.75	-	15	-	-	-
Mixed loads 3/	50	-	-	Per spec		4	-	-	-	4	-	-	-

1/ Absence of value indicates relay is not rated for that application.
2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating.
3/ Mixed loads to be conducted at +71°C. Number of mixed loads cycles following continuous current test - 5000.

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REQUIREMENTS

Weight -001 and -002; 10.6 ounces maximum.
-003, -004, -005, -006, and -007; 11.5 ounces maximum.

Temperature range: -55°C to +71°C.

Maximum altitude rating: 50,000 feet 1/.

Shock g-level: ½ sine, 25 g's.

Duration: 6 to 9 ms.

Maximum duration contact opening: 1 ms.

Vibration-sinusoidal: See table III.

TABLE III. Vibration-sinusoidal.

Dash No.	Vibration level					
					800 - 2000 Hz	
	5-10 Hz	10-55 Hz	55-400 Hz	400-800 Hz	-55°C and 25°C	+71°C
-001, -005, -007			10 g's	10 g's	10 g's	7 g's
-002			10 g's	8 g's	8 g's	7 g's
-003	.08 DA	.06 DA	10 g's	10 g's	10 g's	7 g's
-004			10 g's	8 g's	8 g's	7 g's
-006			10 g's	8 g's	8 g's	7 g's

Vibration (random): Not applicable.

Acceleration: 15 g's/

Seal:

Applicable specification: MIL-STD-202-112, test condition C, procedure IV.

Leak rate: $6x10^{-4}$ STD cc/s.

Strength of terminals and mounting studs:

Torque:

Main terminals: 24 inch-pounds maximum, 18 inch-pounds recommended installation torque.

Auxiliary terminals: 4.4 inch-pounds maximum, 3.3 inch-pounds recommended installation torque.

Insulation resistance, initial: 200 megohms minimum.

After life or environmental tests: 100 megohms minimum.

1/ Application range can be extended to 80,000 feet by suitably encapsulating the terminals. Dielectric withstanding voltage (sea level) 2 to 5 seconds.

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	Initial		After life tests	
	28 V dc	115 V ac	28 V dc	115 V ac
Coil to case -	1250	1500	1000	1125
Auxiliary contacts - - - - - - - - - - - - - - - - -	1250	1250	1000	1000
Across open power contacts - - - - - - - - -	1250	1250	625	625
All other points - - - - - - - - - - - - - - - - - -	1800	1800	1350	1350

Dielectric withstanding voltage (altitude) - 1 minute.

	28 V dc	115 V ac
Coil to case - - - - - - - - - - - - - - - - -	500	500
Auxiliary contacts - - - - - - - - - - - - -	500	500
All other points - - - - - - - - - - - - - - -	700	700

Maximum contact drop, initial .150 volt.

After life test .200 volt.

	NO (V ac)	NC (V ac)
Overload current	320	320
Rupture current	400	400

Coil suppression (transient voltage, back emf): 45V dc maximum.

Duty rating: Continuous.

RFI specification (applicable to coil circuits of ac operated relays): MIL-STD-461.

Part number M6106/9- (dash number from table I).

Qualification by similarity: See MIL-PRF-6106.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

1/ Number of rupture cycles after mixed loads test - 5.

Referenced documents. In addition to MIL-PRF-6106, this document references the following:

MIL-STD-202-112 MIL-STD-461

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Custodians:
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2022-027)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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